

ETHAN KIM

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Chapel Hill, NC

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DoctorKrill08

Portfolio



<https://ethankimufgonline.com>

EDUCATION

University of North Carolina at Chapel Hill

Bachelor of Science Computer Science & Minor in Statistics Expected 2029

Associates in Arts from Guilford Technical Community College

Java Programming Certification from Guilford Technical Community College

SKILLS

Languages: Java, Python, JavaScript/TypeScript, HTML/CSS, Lua

Frameworks & Libraries: FastAPI, React/Next.js, Flask, WebSockets, SocketIO, OpenCV, YOLO

Robotics: Kalman Filters, Control, Mapping, Motion Planning, Inverse Kinematics, Computer Vision

Hardware: NVIDIA Jetson, Arduino, Realsense Depth Cameras, RTK GPS, Lidar

EXPERIENCE

NC A&T CLAEM Internship - Intern (June 2026 - August 2026)

- Developed autonomous navigation software for mobile robots using LIDAR, depth cameras, and RTK GPS.
- Designed Kalman filter sensor fusion integrating RTK GPS, IMU, and wheeled odometry for outdoor localization.
- Combined LiDAR and depth camera perception with localization to create a real-time obstacle map, then used the Dynamic Window Approach for obstacle avoidance.
- Built a full-stack robot control and telemetry interface using FastAPI, React/Next.js, and WebSockets.
- Trained and integrated YOLO object detection models and implemented inverse kinematic control for robot arm use.

PROJECTS

Undertale Fighting Game | Creator & Programmer | March 2022 - Present

- Created and programmed a Roblox multiplayer game with 6M+ visits and 2M+ unique users
- Designed gameplay systems, movement, VFX, interfaces, and everything interactive.

FIRST Tech Challenge (FTC) Team 10195 Night Owls | Captain & Lead Software | May 2024 - June 2026

- Led software development for competitive FTC robots, developing autonomous pathing, PID Control systems, and sensor fusion for localization.
- Used Kalman Filters to combine global AprilTag position estimates and local dead-wheel odometry + IMU position estimates to achieve real-time localization.
- Achieved the **NC State record for Autonomous** Offensive Power Rating in the 2024 - 2025 season and **won the team's first competition** since 2017 in the 2025 - 2026 season.

VOLUNTEERING

YB Champions Taekwondo | Instructor | August 2022 - August 2026

- Led class and instructed students of all ages for over 3 years, 1000+ hours

ECG Robotics Programming Camp | Host/Leader | Summer 2025 and Summer 2026

- Hosted and led a 5 day long programming camp that taught rookie members (from multiple schools, 7th to 10th grade) how to code in Java and some basic robotics concepts.